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An updated and expanded assessment of PLS-SEM in information systems research

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Abstract

Purpose – Following the call for awareness of accepted reporting practices by Ringle, Sarstedt, and Straub in 2012, the purpose of this paper is to review and analyze the use of partial least squares structural equation modeling (PLS-SEM) in *Industrial Management & Data Systems* (*IMDS*) and extend *MIS Quarterly* (*MISQ*) applications to include the period 2012-2014.
Design/methodology/approach – Review of PLS-SEM applications in information systems (IS) studies published in *IMDS* and *MISQ* for the period 2010-2014 identifying a total of 57 articles reporting the use of or commenting on PLS-SEM.
Findings – The results indicate an increased maturity of the IS field in using PLS-SEM for model complexity and formative measures and not just small sample sizes and non-normal data.
Research limitations/implications – Findings demonstrate the continued use and acceptance of PLS-SEM as an accepted research method within IS. PLS-SEM is discussed as the preferred SEM method when the research objective is prediction.
Practical implications – This update on PLS-SEM use and recent developments will help authors to better understand and apply the method. Researchers are encouraged to engage in complete reporting procedures.
Originality/value – Applications of PLS-SEM for exploratory research and theory development are increasing. IS scholars should continue to exercise sound practice by reporting reasons for using PLS-SEM and recognizing its wider applicability for research. Recommended reporting guidelines following Ringle *et al.* (2012) and Gefen *et al.* (2011) are included. Several important methodological updates are included as well.

Keywords PLS, SEM, Information systems, Methods, PLS-SEM, Guidelines

Paper type General review

Introduction

Researchers at all levels seek to use standardized reporting techniques to demonstrate rigor and allow replicability. Information systems (IS) researchers wishing to employ the method of partial least squares structural equation modeling (PLS-SEM) are no different. This method has been widely applied in the IS field (Gefen *et al.*, 2011; Ringle *et al.*, 2012) as well as other disciplines, including marketing (Hair, Sarstedt, Ringle, and Mena, 2012), strategic management (Hair, Sarstedt, Pieper, and Ringle, 2012), international management (Richter *et al.*, 2016), operations management (Peng and Lai, 2012), tourism (do Valle and Assaker, 2015), accounting (Lee *et al.*, 2011), group and organization research (Sosik *et al.*, 2009), and family business (Sarstedt *et al.*, 2014), but does have limitations (Marcoulides *et al.*, 2009). To ensure the method is properly applied and interpreted, it is important that IS researchers are familiar with the context of the current conversation on PLS-SEM applications. Thus, it is useful to obtain a better understanding of how PLS-SEM is being applied in the IS field following the call by Ringle *et al.* (2012) for awareness and application of accepted reporting practices.

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The purpose of this paper is to review and analyze the recent applications of PLS-SEM in selected IS literature from 2010 to 2015. This paper builds on the work of Ringle *et al.* (2012), which examined the use of PLS-SEM in *MIS Quarterly* (*MISQ*). We also include *Industrial Management & Data Systems* (*IMDS*) that publishes a variety of IS studies and has broad appeal and relevance to researchers. During the six-year period, *IMDS* published 58 studies using the PLS-SEM method vs 34 studies by *MISQ*. These findings demonstrate the continued use and acceptance of PLS-SEM as an accepted research method within IS. Before reporting our results, we reviewed recent developments in PLS-SEM.

Recent developments in PLS-SEM

Several software programs are available to execute PLS-SEM. Recent releases of the software include options for executing multi-group analysis (Sarstedt *et al.*, 2011), invariance testing by means of the measurement invariance of composite models (Henseler *et al.*, 2016), linear and non-linear moderation, continuous moderators, confirmatory tetrad analysis (CTA) (Gudergan *et al.*, 2008), and partial least squares prediction-oriented segmentation (PLS-POS) (Becker *et al.*, 2013). Editors and reviewers increasingly are requesting these types of analyses so the ability to easily execute them is a benefit to the researchers.

A somewhat surprising development is proposed approaches referred to as consistent PLS (Bentler and Huang, 2014; Dijkstra, 2014; Dijkstra and Henseler, 2015a, b). The new approaches adapt PLS-SEM to produce the same results as the common factor model of CB-SEM. It is unclear why researchers would use these alternative approaches to PLS-SEM when they could easily apply the much more widely recognized and validated CB-SEM method. Dijkstra and Henseler (2015a, b) note that their approach supposedly corrects the deficiencies of PLS, but that assumes there are in fact deficiencies. Indeed, scholars could just as easily label differences in loadings and path coefficients as deficiencies of CB-SEM since the loadings are in general lower than for PLS-SEM and the coefficients are somewhat higher. Basically, the authors make the same mistake as many CB-SEM scholars when they assume that the common factor model is the benchmark against which PLS-SEM should be compared – a situation referred to as PLS bias or consistency at large, which in fact is not necessarily a bias. It is not surprising that the PLS-SEM method produces parameter estimates that are not the same as CB-SEM, since the algorithms are different and CB-SEM is based only on common variance while PLS is based on total variance (Hair *et al.*, 2016). In sum, these consistent PLS methods are designed for situations in which the research objective is to obtain the same results as CB-SEM, when is very seldom the objective when applying the established PLS-SEM algorithm (Hair *et al.*, 2016).

A final topic to summarize before discussing the results of our analysis of PLS-SEM IS applications is the recent emergence of numerous additional reasons for choosing PLS over CB-SEM. The rules of thumb for selecting each method are listed in Table I. Many methodological developments have emerged for PLS-SEM in recent years and more are on the way. These updates to the PLS-SEM method have extended its analysis capabilities beyond CB-SEM, and as can be noted from Table I, there are quite a few situations where PLS-SEM is the preferred method instead of CB-SEM. Moreover, several of the analysis features of PLS-SEM are not possible with CB-SEM, including continuous moderators, prediction with latent variable scores due to indeterminacy, and higher order constructs with only two first-order constructs. In planning future research, we urge researchers to consider all of the PLS-SEM analysis possibilities when specifying the research design and selecting the method of structural modeling (Table I).

Methodology

Our review of PLS-SEM applications in IS consisted of empirical studies published in *IMDS* and *MISQ* for the period 2012-2015. Papers focusing just on the PLS method and not

Table I.
Rules of thumb for choosing SEM method

PLS-SEM	CB-SEM
1. The research objective is exploratory or confirmation of theory based on total variance	1. The research objective is confirmation of well-developed structural and measurement theory based on common variance
2. The objective of the analysis is prediction	2. The measurement philosophy is estimation with the common factor model using only common variance (covariances)
3. The measurement philosophy is estimation with the composite factor model using total variance	3. The research requires a global goodness-of-fit criterion
4. The research objective is to explain the relationships between exogenous and endogenous constructs	4. The error terms require additional specification, such as covariation.
5. The structural and/or measurement models are complex (many constructs = 6 + and many indicators = 50+)	5. The structural model specifies non-recursive relationships
6. Formatively measured constructs are specified in the research	6. The structural and/or measurement models are simple (5 or fewer constructs and 50 or fewer indicators)
7. Preferred method when sample size is small ($n < 100$). But PLS is also an excellent method for larger samples	
8. The data are not normally distributed	
9. The scaling of responses is ordinal or nominal	
10. The data is secondary/archival, particularly single-item measures	
11. The research objective is to use latent variable scores in subsequent analyses	
12. The structural model will be estimated with a higher order construct that has only two first-order constructs	
13. The analysis involves a continuous moderator	
14. The investigation will examine the model for unobserved heterogeneity	

presenting empirical results were not included in our study. The search identified a total of 92 articles (number of studies/publications) that reported the use of PLS-SEM, or commented on the method (e.g. editorials). Note that findings reported for *MISQ* from prior to 2012 are excerpted from the Ringle *et al.*'s (2012) assessment. This time period was chosen because it reflects the most recent period when the applications of PLS-SEM have grown dramatically, as noted in Figures 1 and 2.

The trends in publishing of articles using PLS-SEM are shown in Figures 1 and 2. The number of PLS-SEM articles in *IMDS* increased considerably over the period, growing from only seven in 2010 to 17 in 2015, for a total of 58. In comparison, a total of 34 PLS-SEM articles were published in *MISQ* for that same period, which was the same as the previous five-year period (2005-2009 = 34). Thus, while the number of PLS-SEM articles in *IMDS* is increasing, the number of similar articles in *MISQ* in recent years appears to be relatively flat. This relatively flat pattern of application of PLS-SEM in *MISQ* may be due to researchers exploring other types of analysis, such as hierarchical linear modeling.

The number of PLS-SEM articles as a proportion of total articles published in *IMDS* from 2010 to 2015 is 13.0 percent (58 of 445), and the proportion for the same years in *MISQ* is 10.9 percent (34 out of 312). Between 2010 and 2015, 29 editorials were included in *MISQ*, and of those editorials four (13.8 percent) discussed PLS-SEM. Additionally, from 1992 to 2015, there were 11.3 percent (88 of 776) PLS-SEM articles published as a proportion of total articles in *MISQ*, and an *MISQ* special issue focused on PLS-SEM in 2009. Thus, overall there are comparable proportions of articles using PLS-SEM in both journals during the relevant periods.

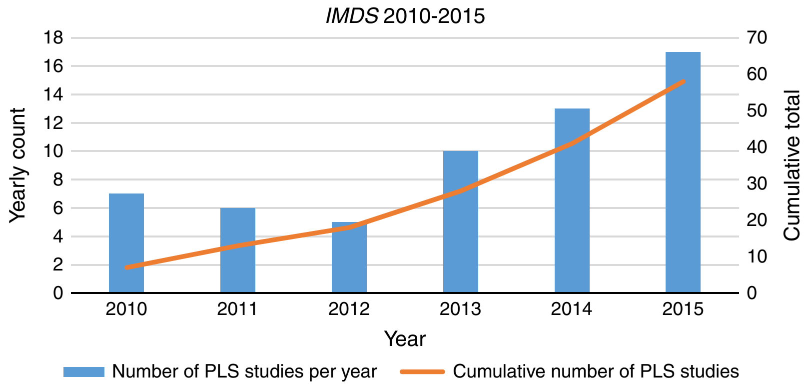

Figure 1. PLS-SEM articles in *IMDS* for the period 2010-2015

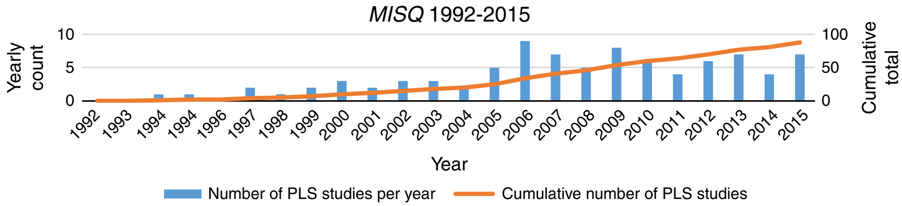

Figure 2. PLS-SEM articles in *MISQ* for the period 1992-2015

Critical issues in PLS-SEM IS applications

Each article that applied PLS-SEM was evaluated according to a wide range of criteria, including reasons for employing PLS-SEM, use of reflective and formative indicators, reliability and validity assessments, multi-group analyses, supplementary analyses, and the software used. The information was systematically recorded in a database for analysis in this study. As a result, we are able to update and expand knowledge about applications of PLS-SEM, as well as trace important developments in the application of PLS-SEM in IS over the past six years. In addition, the analysis enabled us to identify critical issues and typical misapplications of the method. The review focuses on eight critical issues in the application of PLS-SEM identified by Ringle *et al.* (2012), including: reasons given for using PLS-SEM, model descriptive statistics, sampling characteristics, technical reporting, formative measurement metrics, reflective measurement metrics, structural model metrics, and additional analyses such as mediation, moderation, multi-group analyses, and common methods variance. Findings for the two journals are shown to provide a basis of comparison. Where possible, we also suggest best practices on using PLS-SEM as guidelines for future research.

Reasons for using PLS-SEM

For *IMDS*, the most common reason for using PLS-SEM is by far small sample size, with the second most common reason being non-normal data. When compared to *MISQ* PLS-SEM applications, however, model complexity and formative constructs are now the two most common reasons, with exploratory research and small sample size tied the third. It should also be noted that the other category for *MISQ* includes numerous reasons, such as: maximizing explained variance of endogenous variables, number of interaction terms, large number of variables, measures derived from archival data, explains an outcome of interest, identifying relationships, second-order constructs, multiple interdependent relationships, multiple criterion variables, early-stage research, and mediated models. Thus, the dominant reason for using PLS-SEM in *MISQ* is no longer simply small sample size. This seems to indicate that the IS community is heeding recommendations from the Ringle *et al.*'s (2012) assessment and maturing in its use of PLS-SEM.

With regard to *IMDS*, theory development is a widely specified reason, whereas *MISQ* authors specify theory development much less often. *IMDS* also did not have any papers that cited categorical variables, theory testing, or convergence ensured as reasons for selecting PLS-SEM. This lack of specifying these reasons is comparable to the updated *MISQ* counts. Overall, 55 of 58 papers in *IMDS* cite a specific reason for the choice of PLS-SEM, while all 24 of the papers in *MISQ* provide a reason. Thus, continuing to model the practices of one of the leading journals in the field by reporting specific reasons for using PLS-SEM will be helpful for other researchers when they report their work.

A somewhat surprising finding was the low indication of choosing PLS-SEM for prediction. The latent variable scores for CB-SEM are indeterminant – i.e., an infinite number of different sets of latent variable scores that will fit the model equally well are possible for a CB-SEM solution, which makes CB-SEM unsuitable for prediction (Hair *et al.*, 2016, 2018). In contrast, the PLS-SEM method always produces a single determinant score for each SEM composite for each observation. Moreover, CB-SEM prediction as reported in the R^2 is the proportion of common variance explained, whereas PLS-SEM R^2 is the proportion of total variance explained (Hair *et al.*, 2018). Thus, PLS-SEM is always the preferred SEM method when the research objective is prediction and we believe this reason for selecting PLS-SEM rather than CB-SEM will increase considerably in the future (Table II).

Confirmatory or exploratory research

When deciding whether PLS-SEM is the appropriate structural modeling approach, researchers often question whether their research is confirmatory or exploratory research. This is an important concept to understand when analyzing data with SEM since a general rule of thumb for choosing CB-SEM is that it should be used for confirmatory research while

	IMDS 2010-2015		*MISQ* 2012-2015		*MISQ* 1992-2011	
	Number ($n=58$)	Proportion (%)	Number ($n=24$)	Proportion (%)	Number ($n=65$)	Proportion (%)
Total	55	94.8	24	100.0	46	70.8
Specific reasons						
Small sample size	23	39.7	3	12.5	24	36.9
Non-normal data	13	22.4	2	8.3	22	33.9
Formative measures	6	10.3	6	25.0	20	30.8
Focus on prediction	2	3.4	2	8.3	10	15.4
Model complexity	5	8.6	4	16.7	9	13.9
Exploratory research	3	5.2	3	12.5	7	10.8
Theory development	8	13.8	1	4.2	6	9.2
Use of categorical variables	–	–	–	–	4	6.2
Convergence ensured	–	–	–	–	2	3.1
Theory testing	1	–	–	–	1	1.5
Interaction terms	4	6.9	2	8.3	1	1.5
Other/not specified	14	24.1	10	41.7	–	–

Notes: Counts are based on the authors' comments as to their explicit reasons for using PLS-SEM. If they did not specify a reason, it is counted under other/not specified. Some articles specified more than one reason and all reasons are included

Table II.
Reasons for using PLS-SEM

PLS-SEM is preferred for exploratory research, but can also be used for confirmatory research (Hair *et al.*, 2016, 2018).

To clarify this issue, it is useful to clarify how exploratory research differs from confirmatory research. Exploratory research is conducted when problems have not been clearly defined. Researchers may not have enough information to make conceptual distinctions or to propose explanatory relationships, and the approach to the problem must be flexible (Hair *et al.*, 2010). Exploratory research can be used to generate hypotheses from qualitative methods, but it also is used to test hypotheses using quantitative research. For example, when hypotheses are generated by research in another context, e.g., in the USA, you may focus on testing the same or similar hypotheses in another country, e.g., China, or Malaysia. Thus, exploratory research can address all types of research questions, including what, when, why, and how. In contrast, confirmatory research examines previously specified hypotheses that predict specific outcomes based on underlying causal theory, and the hypotheses usually are derived from established causal theories or previous studies conducted within the same context (Hair *et al.*, 2010).

It is useful at this point to clarify the different types of modeling. Predictive modeling is the process of applying a statistical model to examine data with the objective of predicting new or future observations (Shmueli, 2010). This statistical modeling process leads to "statistical conclusions" in terms of explained variance, statistical significance, and effect sizes as they relate to theoretical hypotheses. It should also be noted that in social sciences research the types of statistical models used for testing theoretical causal hypotheses are most often correlation-based models applied to observational/survey data. Thus, statistical modeling is not only predictive modeling, it is also explanatory modeling, or the application of statistical modeling to data to test and explain causal hypotheses about theoretical constructs and structural paths.

In addition to predictive and explanatory modeling, both of which can be accomplished with either CB-SEM or PLS-SEM, a third type of modeling is descriptive. In descriptive modeling there is no underlying causal theory, except in perhaps a limited way (Shmueli, 2010). Therefore, when researchers state they are conducting exploratory research they are representing relationships between data structures, e.g., independent and dependent variables, in a way that summarizes those relationships. In short, they are exploring possible relationships not based on theoretical or causal justification, but rather searching for potential associations that may lead to theory development.

In applying the rule of thumb for selecting the appropriate SEM method, the general definitions of the concepts must be extended to a multivariate statistical context. SEM has two stages – confirmatory factor analysis (CFA) and structural equation modeling. CFA enables the researcher to test the hypothesis that theoretical relationships (sometimes referred to as causal) actually exist between the observed indicator variables and their underlying latent constructs. In short, a CFA assesses measurement theory. Note that when running a CFA with PLS-SEM the process is referred to as confirmatory composite analysis (Henseler *et al.*, 2014). The second SEM stage, structural equation modeling, tests whether the theoretical structural relationships (also sometimes referred to as causal) between the constructs are meaningful and significant. In short, this second stage examines structural theory. The hypothesized measurement and structural model relationships are based on theory and/or previous empirical research, either qualitative or quantitative, and the hypotheses are tested statistically.

CB-SEM and PLS-SEM are both used to conduct a CFA to assess, and perhaps confirm, theoretical measurement theory and structural model relationships. In evaluating the results in the CFA stage, the only difference between the two methods is CB-SEM that is assessed on the basis of reliability, convergent validity, and discriminant validity, as well as on how well the relationships between the indicator variables, as represented by the observed covariance matrix, can be reproduced. The extent to which these relationships can be reproduced is referred to as "goodness-of-fit" (Hair *et al.*, 2010). In contrast, PLS-SEM is not based on covariances and thus does not have a fit measure. The CFA stage in PLS-SEM does, however, test the hypothesis that theoretical relationships actually exist between the observed indicator variables and their underlying latent constructs. With PLS-SEM, to confirm the CFA hypotheses for the measurement model the only metrics applied are reliability, convergent validity, and discriminant validity (Hair *et al.*, 2016, 2018). A similar situation is present when examining the structural model relationships. For both methods, the size and significance of the path coefficients are assessed, but in addition when using CB-SEM researchers must also assess fit.

To summarize, when deciding whether to use CB-SEM or PLS-SEM, researchers should understand that because CB-SEM is based on covariances (only common variance) and requires fit, the method is suitable only for confirmatory research that is based on well-developed theory. In contrast, PLS-SEM is based on total variance and is a useful method for both exploratory and confirmatory research. While PLS-SEM is generally thought of as only for exploratory research, it is also very useful for confirmatory research. The primary difference is that CB-SEM is confirmed with reliability, validity, and goodness-of-fit metrics, whereas PLS-SEM is confirmed with reliability and validity metrics only.

Model descriptive statistics

The average number of latent variables has increased in *MISQ*. *MISQ* articles continue to include models with a large number of variables, with the Ringle *et al.*'s (2012) editorial reporting a high number of 36 latent variables and the updated assessment for 2012-2015 identifying a high number of 25 latent variables. *IMDS* includes primarily reflectively measured constructs, has only two papers reporting both formative and reflective constructs, and none with formative only. *MISQ* has a more balanced modeling mix; i.e., quite a few reflectively measured only papers and also both reflective and formative, but also only a small number of formative only constructs. More recently, there are considerably more model specifications in *MISQ* that are combinations of both formative and reflective constructs.

Additional comparisons show similar numbers of total indicators in the models as numbers of control variables. Much fewer single-item measures are reported in *IMDS* compared to *MISQ*, and this is positive, as recent research raises serious concerns about the validity of such measures (e.g. Diamantopoulos *et al.*, 2012; Sarstedt, Diamantopoulos, Salzberger, and Baumgartner, 2016; Sarstedt, Diamantopoulos, and Salzberger, 2016). Similarly, there are far few higher order constructs than in *MISQ*. Since higher order constructs are becoming more prevalent, we expect their number to increase in *IMDS*. As a best practice, these finding indicate authors should include item wordings, scales, scale means, standard deviations, and a correlation matrix (Table III).

Sampling characteristics

There are differences in sample characteristics between *MISQ* and *IMDS*. Small sample size was indicated less frequently as a primary reason for using PLS-SEM in *MISQ*. Reported sample sizes in *IMDS* do reflect smaller samples are being used in that journal, but the difference in sample sizes is more pronounced in the recent five-year period. In fact, in *IMDS* the "ten times rule" is not always met, whereas in the more recent period for *MISQ* the rule is always met. Recall that the ten times rule indicates the sample size should be equal to the larger of ten times the largest number of formative indicators used to measure a single construct, or ten times the largest number of structural paths directed at a particular construct in the structural model (Hair *et al.*, 2016). With regard to testing for non-response bias, comments are occasionally reported, and range from 27 percent in *IMDS* to 37 percent in *MISQ*. Not surprisingly, validation via a holdout sample is very seldom conducted in either journal, a pattern that is unfortunately typical for most journals (Table IV).

Criterion	*IMDS* 2010-2015 Number ($n=58$)	Proportion (%)	*MISQ* 2012-2015 Number ($n=24$)	Proportion (%)	*MISQ* 1992-2011 Number ($n=109$)	Proportion (%)
Number of latent variables						
Mean	6.72		8.83		8.12	
Median	7	–	8	–	7	–
Range	(2, 12)		(4, 25)		(3, 36)	
Number of structural model relations						
Mean	8.15		11.54		11.38	
Median	8	–	8	–	8	–
Range	(1, 22)		(5, 28)		(2, 64)	
Mode of measurement models						
Only reflective	52	89.7	10	41.7	46	42.2
Only formative	0	0.0	1	4.2	2	1.8
Reflective and formative	6	10.3	12	50.0	33	30.3
Not specified	0	0.0		0.0	28	25.7
Number of indicators per reflective construct[a]						
Mean	11.87		8.75		3.58	
Median	4.9	–	3.69	–	3.5	–
Range	(1, 19)		(1, 11)		(1, 400)	
Number of indicators per formative construct						
Mean	Only 2 included this; 1 and 13 indicators		5.17		3.03	
Median		–	3.3	–	3	–
Range			(2, 7)		(1, 11)	
Total number of indicators in models						
Mean	31.17		33.91		27.42	
Median	29	–	30	–	26.5	–
Range	(13, 65)		(12, 81)		(5, 1,064)	
Number of models with control variables	16		13		29	
Number of control variables						
Mean	2.59	–	6.00		3.69	
Median	3		7		4	
Range	(1, 4)		(3, 17)		(1, 6)	
Number of studies with						
Single-item constructs	6	10.3	11	45.8	31	47.7
Higher order constructs (i.e. hierarchical component analysis	7	12.1	11	45.8	15	23.1
Non-linear relationships)					3	4.6
Model modified in the course of analysis	0	0.0	3	12.5	18	27.7
If yes, comparison with initial model?	0	0.0	3	12.5	6	9.2
Item wordings reported	44	75.9	21	87.5	58	89.2
Scales reported	53	91.4	23	95.8	55	84.6
Scale means and standard deviations reported	23	39.7	20	83.3	43	66.2
Correlation/covariance matrix	52	89.7	20	83.3	54	83.1

Note: [a]$n=56$; two did not report this item

Table III. Model descriptive statistics

	IMDS 2010-2015 Number ($n=58$)	Proportion (%)	*MISQ* 2012-2015 Number ($n=24$)	Proportion (%)	*MISQ* 1992-2011 Number ($n=109$)	Proportion (%)
Sample size						
Mean	309.07		390.71[b]		238.12	
Median	217.5		188[b]		198	
Range	(59, 736)		(128, 1,512)[b]		(17, 1,449)	
Less than 100 observations	9	13.8	0	0	25	22.9
Ten times rule of thumb not met	6[a]	10.3	n/a	0	6	5.5
If not met, to what extent (in percentages) was	35.0				22.5	
The sample size below the required *N*						
According to the ten times rule?						
Non-response bias	18	31.0	9	37.5	24	36.9
Holdout sample used	0	0.0	0	0.0	2	3.1
Missing values reported	3	5.2	0	0.0	10	15.4
Treatment of influential observations (e.g. outliers) reported	3	5.2	1	4.2	4	6.2
Non-normality reported (e.g. skewness, K-S test)	4	6.9	0	0.0	4	6.2

Notes: [a]In total, 56 of the 58 articles included sample size; [b]Sample size of 9,956 was excluded from mean, median, and range calculations as it is an outlier

Table IV. Sampling characteristics

Technical reporting

A fairly high proportion of software users are violating the licensing requirements, which specify that the specific package should be identified. Of the software packages reported, SmartPLS (Ringle *et al.*, 2005) is the most often one identified in *IMDS* but less than 50 percent report it. In contrast, there was a relatively even split between SmartPLS and PLS graph (Chin, 2003), with both packages at about 35 percent in *MISQ* for the most recent period. This suggests an increasing application of SmartPLS compared to the previous study by Ringle *et al.* (2012), where PLS Graph was more widely used. The prevalence of PLS Graph in the earlier *MISQ* period is a result of the fact that SmartPLS was only available from 2005 on, whereas PLS Graph was released in the early 1990s. We expect the proportion of software users to be much higher for SmartPLS in the future, since the current number of analysis features is much greater, but this may change if the PLS Graph developers update that software. Slightly less than 50 percent of articles in *IMDS* comment on bootstrapping, whereas a much higher proportion mentions it in *MISQ*. Editors and reviewers should be urging authors to provide more details, particularly for bootstrapping. Specific metrics to report for bootstrapping include the number of bootstrap samples, significance levels, confidence intervals, and standard errors (Hair *et al.*, 2016) (Table V).

Reported formative measurement metrics

In general, formative measurement is not reported as being used very often in either journal, but it has almost doubled in *MISQ* (32 to 54 percent) since the previous study by Ringle *et al.* (2012). The findings for incorrect assessment of formative measurement models appear to be high, but realistically the sample size is so small that it is risky to interpret it this way. The authors find that quite a few individuals still do not understand that formative measurement models are not

Number of studies reporting	*IMDS* 2010-2015 Number ($n=58$)	Proportion (%)	*MISQ* 2012-2015 Number ($n=24$)	Proportion (%)	*MISQ* 1992-2011 Number ($n=65$)	Proportion (%)
Software used						
PLS Graph (Chin, 2003)	6	10.3	7	29.2	35	53.9
SmartPLS (Ringle *et al.*, 2005)	32	55.2	10	41.7	2	3.1
LVPLS (Lohmöller, 1987)	0	0.0	0	0.0	1	1.5
Not Reported	20	34.5	7	29.2	27	41.5
Resampling method (e.g. bootstrapping)						
Use mentioned	31	53.4	11	45.8	61	93.9
Algorithmic options	0	0.0	0	0.0	24	36.9

Table V. Technical reporting

evaluated based on internal consistency measures such as composite reliability or AVE. The Hair *et al.*'s (2016) book includes a comprehensive coverage of this topic that should be reviewed by individuals not familiar with the differences (Table VI).

Reported reflective measurement model metrics

Authors are reporting reflective measurement model metrics reasonably well. A very high percentage of authors in both journals report indicator loadings. Further, for the 2011-2015 period well over one-half of the studies report both composite reliability and Cronbach's α, and for AVEs the percentage is much higher. Not surprisingly, cross-loadings were reported more in the earlier *MISQ* assessment of discriminant validity, but in the more recent assessment Fornell-Larcker is often the only criterion in *IMDS* while both are typically reported in *MISQ*. This difference may be due to page count limitations where *IMDS* limits authors to 8,000 words (which is estimated at roughly 20 pages) while *MISQ* permits twice that at up to 40 pages for articles. We recommend that authors no longer rely on cross-loadings or even Fornell-Larcker,

Empirical test criterion in PLS-SEM		*IMDS* 2010-2015 Number ($n=4$)	Proportion (%)	*MISQ* 2012-2015 Number ($n=13$)	Proportion (%)	*MISQ* 1992-2011 Number ($n=65$)	Proportion (%)
Reflective criteria used to evaluate formative constructs		4	100.0	5	38.5	5	14.3
Absolute indicator contribution to the construct	Indicator weights	4	100.0	11	84.6	24	36.9
Significance of weights	Standard errors	4	100.0	3	23.1	20	57.1
	Significance levels			8	61.5		
	t-values/*p*-values for indicator weights			10	76.9		
Multicollinearity	Only VIF/tolerance	4	100.0	11	84.6	9	25.7
	Only condition index			0	0.0	0	0.0
	Both			0	0.0	0	0.0

Table VI. Reported formative measurement model metrics as both often substantially overstate the presence of discriminant validity. Instead, authors should report the recently developed HTMT criterion for discriminant validity (Henseler *et al.*, 2015), particularly if space limitations are an issue (Table VII).

Reported structural model metrics

The basic structural model metrics of R^2, path coefficients size, and significance are almost always reported. The other important metrics including f^2, Q^2, and q^2 are much less frequently reported. Authors should always report these metrics to enable accurate interpretation of the results. The small reporting frequency of confidence intervals and total effects in both journals is a result of these metrics being required for only a few types of analyses in previous research. For future research, there are new types of analyses, such as mediation, that will necessitate reporting of these metrics (Table VIII).

Additional considerations and supplementary analyses

For several years, an assessment of the presence of common methods variance has often been required by the editors and reviewers in many journals. When common methods variance assessment has been reported, it typically refers to procedures recommended by Podsakoff *et al.* (2003). Almost 80 percent of articles in *MISQ* comment on common methods variance, whereas only about 44 percent comment in *IMDS*. Assessments of common methods bias may be reported in articles less often in the future, however, as recent research by Fuller *et al.* (2016) and Babin *et al.* (2016) indicate common methods variance is not present nearly as often as suggested in the past (Podsakoff *et al.*, 2003). Moreover, this recent research also concludes that the Harman's (1976) one-factor test is in fact an acceptable method of assessing common methods bias, so when reported the Harman test is likely to be considered as an acceptable assessment tool for common methods variance.

An interesting finding is the increase in applications of mediation in the recent assessment of *MISQ* (35.7 percent). In contrast, the previous *MISQ* assessment reported mediation much less often (23.1 percent), but slightly more than the current applications in *IMDS* (19.7 percent). We expect the application of mediation in PLS studies to increase in all

Empirical test criterion in PLS-SEM		*IMDS* 2010-2015 Number ($n=58$)	Proportion (%)	*MISQ* 2012-2015 Number ($n=24$)	Proportion (%)	*MISQ* 1992-2011 Number ($n=79$)	Proportion (%)
Indicator reliability	Indicator loadings	52	89.7	18	75.0	70	88.6
Internal consistency reliability	Only composite reliability	8	13.8	6	25.0	45	57.0
	Only Cronbach's α	7	12.1	0	0.0	8	10.1
	Both	42	72.4	15	62.5	22	27.9
Convergent validity	AVE 1	53	91.4	20	83.3	70	88.6
	Other	0	0.0	0	0.0	9	11.4
Discriminant validity	Only Fornell-Larcker criterion	42	72.4	8	33.3	29	36.7
	Only cross-loadings	3	5.2	3	12.5	7	8.9
	Both	6	10.3	11	45.8	33	41.8
	Other	0	0.0	0	0.0	3	3.8

Table VII. Reported reflective measurement model metrics

Criterion	Empirical test criterion in PLS-SEM	*IMDS* 2010-2015 Number ($n=61$)	Proportion (%)	*MISQ* 2012-2015 Number ($n=24$)	Proportion (%)	*MISQ* 1992-2011 Number ($n=109$)	Proportion (%)
Coefficient of determination	R^2	58	95.1	22	91.7	105	96.3
	f^2 effect size	10	16.4	13	54.2	13	11.9
Predictive relevance	Cross-validated	0	0.0	1	4.2	0	0.0
	Redundancy Q^2	14	23.0	0	0.0		
	q^2 effect size	4	6.6	1	4.2	0	0.0
Path coefficients	Absolute values	48	78.7	22	91.7	107	98.2
Significance of path coefficients	Standard errors, significance levels, *t*-values, *p*-values	56	91.8	22	91.7	107	98.2
Confidence intervals	–	0	0.0	2	8.3	0	0.0
Total effects	–	3	4.9	2	8.3	4	3.7

Table VIII. Reported structural model metrics

IS journals as editors and reviewers are increasingly requesting it. In addition, the PLS-SEM approach to examining mediation overcomes previous shortcomings of the Baron and Kenny's (1986) method, noted by Zhao *et al.* (2010), and is superior to the Preacher and Hayes' (2008) process approach that relies on multiple regression.

There are quite a few other analyses that should be considered for future applications of PLS-SEM. One of the primary issues to focus on is the possible presence of unobserved heterogeneity, which if overlooked can be a threat to the validity of PLS-SEM findings. Observed heterogeneity is a situation in which on an a priori basis potential characteristics of subgroups, such as male or female, or country of origin, are known and can be examined. In contrast, unobserved heterogeneity is the opposite situation in which there are unobservable characteristics that cause differences in subgroups and thus the theoretical model cannot be examined as a single homogeneous population. To identify unobserved heterogeneity, there are multiple approaches generally referred to as latent class techniques. We suggest you to refer to the most recent discussions of this topic that summarize the benefits of combining FIMIX-PLS and PLS-POS (e.g. Sarstedt, Ringle and Gudergan, 2016). Another topic that may be useful for researchers is confirmatory tetrad analysis in PLS-SEM (CTA-PLS) (Gudergan *et al.*, 2008). CTA-PLS is a method of empirically testing and evaluating the cause-effect relationships for latent variables as well as the specification of indicators in measurement models (Hair *et al.*, 2016). This test assists in avoiding misspecification of formative and reflective indicators (Table IX).

Best practices recommendations on reporting PLS-SEM results

PLS-SEM is a widely applied tool in the IS literature. Its attractiveness for IS scholars and practitioners can be attributed to several characteristics. First, as a limited information approach (Dijkstra, 1983), there are few assumptions about the population or scale of measurement (Fornell and Bookstein, 1982) and therefore nominal, ordinal, interval, and ratio scaled variables can be used when interpreted according to guidelines for this type of measurement. Second, PLS-SEM achieves high levels of statistical power with small sample sizes and complex models (Chin and Newsted, 1999; Reinartz *et al.*, 2009). The general PLS path modeling algorithm is based on ordinary least squares regression for separate subparts of the path model. Therefore, the complexity of the overall model has minimal influence on sample size requirements. Third, PLS-SEM is preferred over CB-SEM in many

Table IX.
Additional considerations and supplementary analyses

Criterion	*IMDS* 2010-2015 Number ($n=61$)	Proportion (%)	*MISQ* 2012-2015 Number ($n=24$)	Proportion (%)	*MISQ* 1992-2011 Number ($n=65$)	Proportion (%)
Common method variance	27	44.3	19	79.2	26	40.0
Mediator analysis	12	19.7	9	37.5	15	23.1
Multi-group analysis						
Continuous moderator analysis	0	0.0	2	8.3	8	12.3
Categorical, observed (multi-group comparison)	2	3.3	0	0.0	16	24.6
Categorical, unobserved (model-based)	0	0.0	0	0.0	0	0.0
Segmentation techniques; e.g., FIMIX-PLS)						
Measurement model invariance	0	0.0	0	0.0	3	4.6
Tetrad analysis	0	0.0	0	0.0	1	1.5

research contexts (see Table I), particularly when the statistical objective is prediction. Fourth, PLS-SEM readily incorporates constructs that are measured either reflectively or formatively, making the approach particularly appealing for success factor studies (Albers, 2010). Finally, user-friendly software with a graphical user-interface and many options for advanced analyses, like SmartPLS (Ringle *et al.*, 2015) and PLS Graph (Chin, 2003), have contributed to the attractiveness of PLS-SEM.

Our review of PLS-SEM applications in IS suggests that PLS-SEM's methodological properties continue to be misunderstood, and at times this has led to misapplication of the technique. Table X includes guidelines for best practices in the application of the PLS-SEM to IS research.

Conclusions

This paper heeds the call for awareness and application of accepted reporting practices by Ringle *et al.* (2012). It reviews and analyzes the use of PLS-SEM from 2010 to 2015 in two representative journals for the field of IS: *IMDS* and *MISQ*. Findings indicate an increased maturity in the application of PLS-SEM by IS researchers. Specifically, PLS-SEM is being chosen less often due to small sample sizes and non-normal data, and much more often as a result of model complexity and the use of formative measurement approaches. At the same time, exploratory research and theory development are increasingly listed as reasons for choosing PLS-SEM, particularly in *IMDS*. The IS field should continue to exercise sound practice by improved reporting of PLS-SEM assessment metrics and recognizing its wider applicability for research. Future studies should expand this assessment by examining the application of PLS-SEM in other journals and including metrics on the recently emerging analyses, such as mediation, moderation, invariance, unobserved heterogeneity, and multi-group analysis.

There are many new options for further PLS-SEM analyses that will lead to better understanding of data relationships. These new options provide additional reasons to apply PLS to explain SEM relationships. The traditional CB-SEM approach has several unique situations where it is the preferred SEM approach. Overall, however, PLS-SEM has many situations where it is the preferred SEM approach. And a number of these situations, such as continuous moderators and higher order constructs with only two first-order constructs, cannot be executed using the CB-SEM method.

As a final thought, authors should remember that PLS-SEM and CB-SEM differ from a statistical point of view, are designed to achieve different objectives, and rely on different

Criterion	Rules of thumb
Characteristics of the data	
Sample size	Use "ten times rule" as rough guideline for minimum sample size; adjust sample size considering power
Sample distribution	Very robust when applied to highly skewed data
Use of holdout sample	Minimum of 30% of original sample
Number of missing values	PLS-SEM is robust with approximately 10% missing at random data
Missing value treatment	Do not use mean replacement; delete case-wise or carry out nearest neighbor or EM algorithm imputation; code missing data where appropriate
Measurement scales	All types OK, except categorical scale for ultimate dependent variable
Algorithm settings and software	
Starting values of weights for initial approximation of latent variable scores	Use 1 as an initial value for each outer weight
Weighting scheme	Use path weighting scheme
Stop criterion	Sum of the outer weights' changes between two iterations $< 10^{-5}$
Software	Report full citation as required by license agreements
Parameter settings	
Bootstrapping sign changes option	Use individual sign changes
Number of bootstrap samples	5,000; must be greater than the number of observations
Size of bootstrap samples	Equal to the number of observations
Blindfolding	Use cross-validated redundancy
Omission distance d	Number of valid observations divided by d must not be an integer; $5 \leqslant d \leqslant 10$
Multi-group comparisons	Use non-parametric approach to assess significance
Model characteristics	
Inner model description	Structural model should display all inner model relationships
Outer model description	A list of all measurement model indicators (questions)
Latent variable measurement mode	Endogenous latent variables should not be modeled formatively; evaluate measurement mode with CTA-PLS (confirmatory tetrad analysis)
Number of items on latent variable	In general avoid single-item measures
Outer model evaluation: reflective measurement models	
Indicator loadings size	Standardized indicator loadings $\geqslant 0.70$
Construct reliability	Composite reliability $\geqslant 0.70$ (in exploratory research 0.60 to 0.70 is considered acceptable)
Convergent validity	AVE $\geqslant 0.50$
Discriminant validity – HTMT	Values lower than 0.85 for conceptually distinct constructs and below 0.90 for conceptually similar constructs; confidence intervals should not include a value of 1
Measurement model specification	Consider executing CTA-PLS to empirically evaluate model specification
Outer model evaluation: formative measurement models	
Significance of weights	Report t-values and p-values
Multicollinearity	Examine for VIF < 5
Absolute vs relative indicator contributions	Report indicator weights and loadings; assess significance
Inner model evaluation	
R^2	Research context determines acceptable level
Effect size f^2	0.02, 0.15, 0.35 for weak, moderate, strong effects
Path coefficient estimates	Assess significance and confidence intervals

(*continued*)

Table X. Best practices: reporting PLS-SEM results

Criterion	Rules of thumb
Predictive relevance Q^2 and q^2	Use blindfolding; q^2: 0.02, 0.15, 0.35 for weak, moderate, strong predictive relevance
Observed and unobserved heterogeneity	Consider categorical or continuous moderating variables using a priori information, FIMIX-PLS or PLS-POS
Other	
Multi-group analysis	When comparing groups specified with observed or unobserved heterogeneity, group sizes should be comparable. When group sizes are quite different (e.g. ratio ⩾ 2:1), randomly withdraw observations from larger group so group sizes are comparable

Table X.

measurement philosophies – total variance for PLS-SEM vs common variance only for CB-SEM. Overall, neither of the methods is generally superior to the other and neither method is appropriate for all situations. In general, the strengths of PLS-SEM are CB-SEM's limitations, and vice versa. It is important that researchers be aware of the different applications for which each approach was developed and apply the methods accordingly.

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